

| Question 1 |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |            |          |          | Maths    | Prac     |
|------------|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|----------|----------|----------|----------|
|            |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2        | AO3      | Total    |          |          |
| 1          | (a) | (i)  | Conversion of eV to J i.e. any 1 of:<br>$0.820\text{ eV} = 1.312 \times 10^{-19}\text{ J}$<br>$0.051\text{ eV} = 0.082 \times 10^{-19}\text{ J}$<br>$0.769\text{ eV} = 1.23 \times 10^{-19}\text{ J}$ or by implication (1)<br>Photon energy = $0.82\text{ eV} - 0.05\text{ eV}$ <b>or</b> $1.312 \times 10^{-19}\text{ J} - 0.082 \times 10^{-19}\text{ J} = [1.230 \times 10^{-19}\text{ J}]$ <b>or</b> by implication (1)<br>$\lambda = 1\,617\text{ [nm]}$ (1)<br>[1516 nm arises from wrong photon energy – 1 mark]<br>Arithmetic slip lose 1 mark | 1               | 1<br><br>1 |          | 3        | 3        |          |
|            |     | (ii) | A photon of this wavelength or energy or $1.23 \times 10^{-19}\text{ J}$ or $0.769\text{ eV}$ or $1\,617\text{ nm}$ ( <b>ecf</b> ) (accept "correct energy, specific energy") passing an electron in <u>the U state</u> (1) accept "colliding with"<br>... triggers release of another photon (1) don't accept "photon absorbed and emits two"<br>The photons are in phase <b>or</b> of the same wavelength <b>or</b> energy <b>or</b> frequency <b>or</b> identical <b>or</b> travelling in the same direction (1)<br>don't accept coherent            | 3               |            |          | 3        |          |          |
|            | (b) |      | Level L self-emptying or short-lived or short time or fall immediately or equivalent (1)<br>Population inversion easier to achieve (accept better or equivalent) <b>or</b> lower pumping [rate] needed (1) don't accept more efficient only                                                                                                                                                                                                                                                                                                             | 2               |            |          | 2        |          |          |
|            |     |      | <b>Question 1 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>6</b>        | <b>2</b>   | <b>0</b> | <b>8</b> | <b>3</b> | <b>0</b> |